IEOR E4004 Optimization Models and Methods, Section 001

Milestone 3: Parameter Estimation & Draft Model

Optimization for a Northeast Corridor Natural Gas Merchant

Team 23 (L) • October 21, 2026

Summary

A natural gas merchant, modeled on BP's U.S. gas marketing business, buys gas at a western Pennsylvania hub and ships it on firm pipeline capacity to contracted customers in New York, New Jersey, and eastern Pennsylvania. It leases storage at the hub and in New York, and can also buy gas at each market's delivered (citygate) price. A multi-period linear program chooses monthly purchases, storage operations, shipments, spot purchases, and firm capacity over the storage year (April 2025 – March 2026) to meet demand at minimum cost. The core trade-off is how to cover winter demand: firm pipeline capacity, summer fills of market-area storage, or winter spot purchases.

Sets

- $T = \{1, \ldots, 12\}$: months, April 2025 ($t = 1$) through March 2026 ($t = 12$)
- $J = \{\text{NY}, \text{NJ}, \text{PA}\}$: delivery markets
- $L = \{1, 2, 3, 4\}$: firm pipeline routes from the hub; $L_j \subseteq L$ are the routes into market j: $L_{\text{NY}} = \{1, 2\}$, $L_{\text{NJ}} = \{3\}$, $L_{\text{PA}} = \{4\}$
- $F = \{H, M\}$: leased storage fields at the hub ($F_0 = \{H\}$) and in New York ($F_{\text{NY}} = \{M\}$; $F_{\text{NJ}} = F_{\text{PA}} = \emptyset$)

Decision Variables

- x_t: gas purchased at the hub in month t (Dth)
- $q^{\text{in}}_{f,t}$, $q^{\text{out}}_{f,t}$: gas injected into and withdrawn from field f in month t (Dth)
- $I_{f,t}$: inventory in field f at the end of month t (Dth)
- $s_{\ell,t}$: gas shipped on route ℓ in month t, measured at the hub (Dth)
- $u_{j,t}$: gas bought at the citygate price in market j in month t (Dth)
- k_ℓ: firm capacity contracted on route ℓ (Dth/day)

All decision variables are continuous and non-negative.

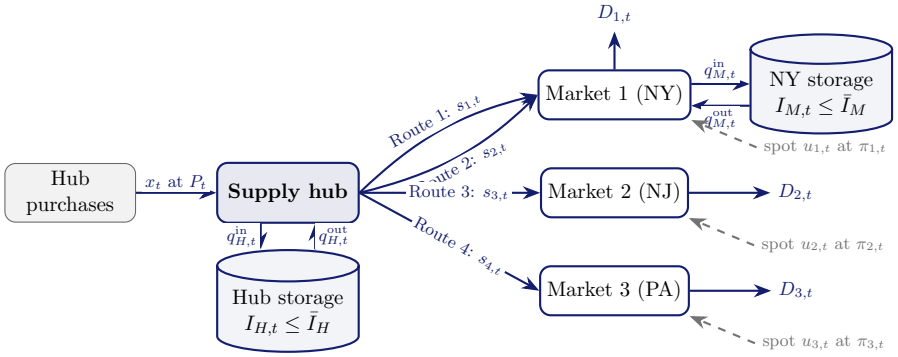


Figure 1: The merchant's network. Hub purchases and hub-storage withdrawals share the four pipeline routes; New York is served by two competing routes and a market-area storage field; every market can also buy at its citygate price (dashed).

Objective Function

Minimize total annual cost: hub purchases, pipeline usage, storage injection and withdrawal, spot purchases, and pipeline reservation charges.

$$\min \sum_{t\in T}\Big[P_t x_t + \sum_{\ell\in L} c_\ell s_{\ell,t} + \sum_{f\in F}\big(c_f^{\text{in}} q_{f,t}^{\text{in}} + c_f^{\text{out}} q_{f,t}^{\text{out}}\big) + \sum_{j\in J}\pi_{j,t}u_{j,t}\Big] + 12\sum_{\ell\in L} R_\ell k_\ell$$

P_t = hub price; c_ℓ = pipeline usage charge; $c_f^{\text{in}}, c_f^{\text{out}}$ = storage injection and withdrawal charges; $\pi_{j,t}$ = citygate price (all \$/Dth). R_ℓ = monthly reservation charge (\$ per Dth/day of capacity per month).

Because customer volumes and prices are fixed by contract (A1), revenue does not depend on these decisions, so minimizing supply cost maximizes the merchant's margin.

Constraints

– **(HB) Hub balance.** Hub purchases plus net withdrawals from hub storage are shipped on the pipeline routes.

$$x_t + \sum_{f\in F_0}\big(q_{f,t}^{\text{out}} - q_{f,t}^{\text{in}}\big) = \sum_{\ell\in L} s_{\ell,t} \qquad \forall\, t\in T$$

– **(DM) Market balance.** Pipeline deliveries net of fuel, plus net withdrawals from storage in the market, plus spot purchases, meet contracted demand $D_{j,t}$.

$$\sum_{\ell\in L_j}(1-\phi_\ell)\, s_{\ell,t} + \sum_{f\in F_j}\big(q_{f,t}^{\text{out}} - q_{f,t}^{\text{in}}\big) + u_{j,t} = D_{j,t} \qquad \forall\, j\in J,\ t\in T$$

ϕ_ℓ = fuel retention rate (fraction of gas shipped).

Gas injected into New York storage must first cross the pipeline (or be bought locally), so summer capacity that would otherwise sit idle can serve winter demand. Market-area storage and firm winter capacity are therefore partial substitutes.

– **(IB) Inventory balance.** Inventory carries over, plus injections minus withdrawals.

$$I_{f,t} = I_{f,t-1} + q_{f,t}^{\text{in}} - q_{f,t}^{\text{out}} \qquad \forall\, f\in F,\ t\in T$$

$I_{f,0}$ = inventory at the start of April 2025.

– **(PC) Pipeline capacity.** Monthly shipments cannot exceed contracted capacity, which cannot exceed the capacity available to the merchant.

$$s_{\ell,t} \le d_t\, k_\ell, \qquad k_\ell \le \bar{k}_\ell \qquad \forall\, \ell\in L,\ t\in T$$

d_t = days in month t; $\bar{k}_\ell = \beta_\ell K_\ell$, where K_ℓ is the pipeline's full capacity and β_ℓ the merchant's accessible share (Dth/day).

– **(SL) Storage limits.** Injection, withdrawal, and inventory stay within field limits.

$$q_{f,t}^{\text{in}} \le d_t\, r_f^{\text{in}}, \qquad q_{f,t}^{\text{out}} \le d_t\, r_f^{\text{out}}, \qquad I_{f,t} \le \bar{I}_f \qquad \forall\, f\in F,\ t\in T$$

$r_f^{\text{in}}, r_f^{\text{out}}$ = maximum daily injection and withdrawal (Dth/day); $\bar{I}_f$ = leased working-gas capacity (Dth).

– **(CI) Cyclic inventory.** Each field ends the year at its starting inventory, so stored gas is not drawn down for free.

$$I_{f,12} = I_{f,0} \qquad \forall\, f\in F$$

Parameter Estimation

All volumes are in Dth and prices in \$/Dth (1 Dth = 1 MMBtu); EIA figures in Mcf are converted at 1.037 Dth/Mcf, EIA's average heat content. All inputs cover April 2025 – March 2026, and monthly values are stored in CSV files read by the Gurobi model.

Demand

Markets are defined at the state level to match EIA's monthly delivery series [1]. We size the merchant against NJR Energy Services, which sold about 108.6 Bcf in FY2025 [2], and assume a 30 Bcf Northeast book split 45% NY, 25% NJ, 30% PA. Dividing by each state's annual deliveries gives merchant shares α_j; applying them to monthly deliveries preserves the seasonal profile. BP Energy Company transactions in Con Edison and New Jersey Natural Gas filings [8] confirm the order of magnitude.

Parameter	Dim.	Source	Method	Value / range
α_j	$\lvert J\rvert$	NJR FY2025 [2]; EIA deliveries [1]	Assumed book ÷ annual state deliveries	NY 0.975%, NJ 1.046%, PA 0.538%
$D_{j,t}$	$\lvert J\rvert \times 12$	EIA monthly deliveries by state [1]	$\alpha_j\times$ deliveries; five missing months imputed from historical year-on-year change	Total 1.68M (Sept.) – 4.22M (Jan.)

A1. Fixed contract revenue. Volumes and sale prices are set by contract and surplus gas is not resold.

A2. Book size is a scenario. The 30 Bcf book and state split are assumptions, not reported market shares.

Storage

Each field is a leased share of its state's working-gas capacity [1] (PA 426.3 Bcf for H, NY 125.2 Bcf for M), with starting inventory set by each state's March 2025 fill ratio.

Parameter	Dim.	Source	Method	Value / range
$\bar{I}_f$	$\lvert F\rvert$	EIA working-gas capacity [1]	Leased share: 0.5% (PA), 1.0% (NY)	H 2.21M; M 1.30M
$I_{f,0}$	$\lvert F\rvert$	EIA inventory, Mar. 2025 [1]	Fill ratio $\times\bar{I}_f$ (PA 29.7%, NY 33.6%)	H 0.66M; M 0.44M
$c_f^{\text{in}},\ c_f^{\text{out}}$	$\lvert F\rvert$	National Fuel FSS [5]	Same benchmark fee for both fields	0.0381 each (\$/Dth)
$r_f^{\text{in}},\ r_f^{\text{out}}$	$\lvert F\rvert$	UGI; Crestwood; EIA [6, 7, 1]	Scaled daily rates; constant limits (Dth/day)	H: 12,280 / 50,054; M: 12,983 / 25,966 (in/out)

Rates are scaled estimates. The common storage fee is a benchmark; additional surcharges and inventory-dependent rate changes are omitted.

A3. Leased shares are scenarios. The 0.5% and 1.0% shares (3.4 Bcf, 11% of annual demand) are not observed and will be varied in the sensitivity analysis.

A4. Market-area storage is stylized. M represents storage downstream of the NY pipeline bottleneck, accessed through utility asset management agreements.

A5. Lease fees are sunk. With $\bar{I}_f$ fixed, lease fees do not affect decisions; pricing storage capacity as a decision is a planned refinement.

Prices

The hub is Eastern Gas South (formerly Dominion South), the most liquid Appalachian pricing point. Citygate prices with missing months (PA Aug. 2025 and Feb. 2026, NY Mar. 2026) are linearly interpolated.

Parameter	Dim.	Source	Method	Value / range
P_t	12	Eastern Gas South monthly index [3]	2025–26 actuals	1.62 (Oct.) – 7.15 (Feb.)
$\pi_{j,t}$	$\lvert J\rvert \times 12$	EIA citygate price by state [1]	2025–26 actuals, converted to \$/Dth	NY 4.53 – 10.03; NJ 4.36 – 7.13; PA 4.66 – 7.22

A6. Price-taker. The merchant buys any quantity at P_t or $\pi_{j,t}$ without moving the price.

A7. Citygate as delivered spot price. The EIA citygate price is a monthly utility average, not a tradable index, so cold-snap price spikes are averaged out; this likely understates the value of winter capacity and storage.

A8. Deterministic inputs. Prices and demand come from the same year, preserving winter co-movement, and are treated as known; uncertainty is examined later through scenarios.

Pipelines

Routes 1–3 are Empire Pipeline and National Fuel Gas Supply into New York and Tennessee Gas Pipeline into New Jersey; Route 4 is Texas Eastern Transmission into Pennsylvania. All are priced at their 2025–26 tariff rates [4]; Empire's reservation charge is month-weighted across its November 1 rate change.

Merchant capacity share. Analogous to α_j, the share β_ℓ is set so that firm capacity into each market equals that market's average daily demand, split equally across its routes: NY 38,355 Dth/day, NJ 21,308, PA 25,570. Firm transport thus covers baseload, and winter demand above it (NY's January monthly average reaches 64,351 Dth/day) must come from storage or spot purchases.

Parameter	Dim.	Source	Method	Value / range
c_ℓ	$\lvert L\rvert$	Pipeline tariffs [4]	Usage charge	R1 0.0009; R2 0.0069; R3 0.0539; R4 0.1181
R_ℓ	$\lvert L\rvert$	Pipeline tariffs [4]	Month-weighted reservation charge (\$/Dth/day/month)	R1 7.00; R2 6.14; R3 4.67; R4 15.63
K_ℓ	$\lvert L\rvert$	Pipeline tariffs [4]	Full firm capacity (Dth/day)	R1 674,050; R2 643,977; R3 1,644,682; R4 453,103
β_ℓ	$\lvert L\rvert$	Derived	Avg. daily market demand ÷ ($\lvert L_j\rvert \times K_\ell$)	R1 2.85%; R2 2.98%; R3 1.30%; R4 5.64%
$\bar{k}_\ell$	$\lvert L\rvert$	Derived	$\beta_\ell K_\ell$ (Dth/day)	R1–R2 19,178; R3 21,308; R4 25,570
ϕ_ℓ	$\lvert L\rvert$	Pipeline tariffs [4]	Deferred (A11)	0

A9. Accessible capacity is a scenario. Access is assumed through capacity release or asset management agreements. Baseline shares range from 1.30% (Tennessee) to 5.64% (Texas Eastern). In the sensitivity analysis we scale every $\bar{k}_\ell$ from 0.5× to 3× its baseline; about 1.7× lets the NY routes cover January's average demand.

A10. Simple route structure. Annual routes link the hub directly to each market. Seasonal capacity, capacity-release trading, interconnections, and in-state distribution are omitted.

A11. Fuel retention deferred. $\phi_\ell = 0$ initially; retention may be added later.

A12. Monthly aggregation. Daily limits are scaled by d_t, ignoring within-month peaks.

References

[1] U.S. Energy Information Administration, Natural Gas Data: state deliveries, citygate prices, underground storage capacity and inventories, and field-level storage. https://www.eia.gov/naturalgas/data.php

[2] New Jersey Resources Corporation, *Form 10-K*, fiscal year ended September 30, 2025.

[3] S&P Global Commodity Insights, *Platts Gas Daily*, Eastern Gas South monthly index (via NSAI).

[4] Federal Energy Regulatory Commission, eTariff (Empire Pipeline, National Fuel Gas Supply, Tennessee Gas Pipeline). https://etariff.ferc.gov

[5] National Fuel Gas Supply Corporation, *Open Season OS370*, September 2025, p. 4 (FSS commodity injection/withdrawal charge). https://www.nationalfuel.com/wp-content/uploads/documents/370.pdf

[6] UGI Storage, *Service Offerings*, Tioga-Mansfield injection capability. https://ugistorage.com/about-us/service-offerings/

[7] Crestwood Equity Partners LP, *Form 10-K*, year ended December 31, 2020, Thomas Corners storage capacity and operating rates. https://www.sec.gov/Archives/edgar/data/1136352/000113635221000003/ceqp-20201231.htm

[8] Consolidated Edison Company of New York and New Jersey Natural Gas Company, utility filings, 2025.